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ITAI 3377: Artificial Intelligence for Edge and IoT Devices
A01: Understanding Development Environments and Tools for Edge AI and IoT
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Edge AI and IoT Development Tools Report

A. Visual Studio Code (VS Code)

Description: VS Code is a free Microsoft code editor made by that helps you write, fix, and manage your code with built-in tools and add-ons.

Purpose: It provides a quick, responsive workspace that supports a wide range of programming languages and offers a large library of extensions. For example, you could install an extension to write Python code for a Raspberry Pi project and another to manage your files on GitHub, all in one place.

Typical Use Cases:

- Writes and debugs Python scripts for TensorFlow Lite models deployed on edge devices, such as building a voice recognition script for a smart home device and stepping through the code to fix errors
- Developing firmware for IoT microcontrollers using the PlatformIO extension, such as writing C++ code for an ESP32 running a TinyML motion detection model while using the integrated debugger to troubleshoot sensor data collection
- Managing Edge Impulse projects through the integrated terminal and version control, such as pushing model updates to GitHub and running deployment commands without leaving the editor

B. Node.js

Description: Node.js is a free, open-source platform that allows JavaScript to run on servers instead of just in browsers. It's designed to handle multiple connections at the same time without slowing down.

Purpose: Its purpose is to let developers create quick, scalable network applications that manage multiple connections at the same time. For instance, it can process live data streams from dozens of smart devices without slowing down.

Typical Use Cases:

- Building MQTT brokers and clients for IoT device communication, such as a smart factory collecting vibration sensor data from multiple machines to detect anomalies in real-time
- Creating REST APIs that aggregate sensor data from edge devices
- Developing real-time dashboards that display live telemetry from IoT networks and push alerts to monitoring systems

C. Edge Impulse CLI

Description: Edge Impulse CLI is a text-based tool that helps developers collect data from sensors, create machine learning models, and install them on small devices without needing a graphical interface.

Purpose: Its purpose is to make the machine learning process easier for small devices by handling data collection, model training, and deployment all in one place. For instance, you can gather sensor data, build a model, and load it onto a microcontroller without switching tools.

Typical Use Cases:

- Capturing audio samples from a microcontroller for keyword spotting model training
- Deploying optimized neural networks to Arduino or Raspberry Pi devices
- Connecting devices like the Nordic Thingy:53 to collect accelerometer data for gesture recognition, training a model in the cloud, and flashing the optimized model back to the device through terminal commands

D. TensorFlow and TensorFlow Lite

Description: TensorFlow is a free machine learning platform created by Google that helps developers build and train AI models. TensorFlow Lite is a compact, efficient version made specifically for running on mobile phones and low-power devices.

Purpose: Its purpose is to let developers build and test machine learning models, then run them on small, low-power devices. For instance, you could train an image recognition model on a computer and then deploy a lighter version to a Raspberry Pi.

Typical Use Cases:

- Training image classification models in TensorFlow, then converting them to TFLite for smartphone or edge deployment

- Running object detection on Raspberry Pi using TensorFlow Lite with hardware acceleration
- Deploying wildlife conservation models, such as training a bird species classifier on cloud GPUs and running it on solar-powered camera traps that identify species without internet connectivity

E. Google Colab

Description: Google Colab is a free online coding platform that runs in your browser. It offers powerful computing resources like GPUs and TPUs, comes with common libraries already installed, and allows easy collaboration—no need to install anything on your computer.

Purpose: Its purpose is to let developers train machine learning models using free cloud computing power without needing their own expensive hardware. For example, a student could build and test an image classifier using a cloud GPU at no cost.

Typical Use Cases:

- Training deep learning models on free GPU resources before converting to edge-compatible formats
- Collaboratively developing AI experiments, such as a research team training a plant disease detection model on thousands of images using free TPU access before exporting for agricultural drone deployment
- Prototyping and testing TensorFlow Lite model conversions before deployment to embedded devices

F. Generative AI Coding Tools (GitHub Copilot, OpenAI Codex)

Description: Generative AI coding tools are AI-powered assistants that suggest code completions, generate functions, explain code, and help fix errors in real-time within your coding environment.

Purpose: Their purpose is to speed up coding by handling repetitive tasks and offering smart suggestions as you type. For example, they can automatically write common code patterns so developers can focus on more complex problems.

Typical Use Cases:

- Auto-generating boilerplate code for IoT sensor initialization and data parsing
- Getting suggestions for TensorFlow model architecture based on natural language descriptions

- Building IoT applications like a smart thermostat by using AI to quickly generate MQTT connection code, temperature sensor reading functions, and TensorFlow Lite inference routines from descriptive comments